Jordan's Lemma

Source: Standard complex-analysis result; treatment follows J. E. Santos' Cambridge IB lectures.

Statement. Let $f(z) = e^{iaz}g(z)$ with $a > 0$. Show that if $g(z) \to 0$ uniformly on the upper half-plane as $|z| \to \infty$, i.e.

$$\max_{\theta\in[0,\pi]} |g(Re^{i\theta})| \to 0 \quad \text{as } R \to \infty, \tag{1.1}$$

then the integral over the upper semicircular arc $\mathcal{C}_R = \{z = Re^{i\theta}, \theta \in [0, \pi]\}$ vanishes:

$$\lim_{R\to\infty} \int_{\mathcal{C}_R} f(z)\, dz = 0. \tag{1.2}$$

Find an upper bound for $\left|\int_{\mathcal{C}_R} f(z)\, dz\right|$ and give an example of a class of functions g satisfying (1.1).

Solution. A typical example satisfying (1.1) is any rational function $g(z) = P(z)/Q(z)$ with $\deg Q \geq \deg P + 1$, for which $\max |g(Re^{i\theta})| = \mathcal{O}(R^{-1}) \to 0$.

Parametrise $z = Re^{i\theta}$, so $dz = iRe^{i\theta}d\theta$. The integral becomes

$$\int_{\mathcal{C}_R} f(z)\, dz = iR \int_0^\pi g(Re^{i\theta})\, e^{iaRe^{i\theta}}\, e^{i\theta}\, d\theta. \tag{1.3}$$

Taking the absolute value and using $|e^{i\theta}| = 1$ and $|e^{iaRe^{i\theta}}| = e^{-aR\sin\theta}$:

$$\begin{aligned} \left|\int_{\mathcal{C}_R} f(z)\, dz\right| &\leq R \int_0^\pi |g(Re^{i\theta})|\, e^{-aR\sin\theta}\, d\theta \\ &\leq M(R) \cdot R \int_0^\pi e^{-aR\sin\theta}\, d\theta, \end{aligned} \tag{1.4}$$

where $M(R) \equiv \max_{\theta\in[0,\pi]} |g(Re^{i\theta})|$. By symmetry of $\sin\theta$ about $\theta = \pi/2$:

$$R \int_0^\pi e^{-aR\sin\theta}\, d\theta = 2R \int_0^{\pi/2} e^{-aR\sin\theta}\, d\theta. \tag{1.5}$$

Since $\sin\theta$ is concave on $[0, \pi/2]$, it lies above the chord connecting $(0, 0)$ and $(\pi/2, 1)$:

$$\sin\theta \geq \frac{2\theta}{\pi}, \qquad \theta \in \left[0, \tfrac{\pi}{2}\right]. \tag{1.6}$$

Therefore:

$$2R \int_0^{\pi/2} e^{-aR\sin\theta}\, d\theta \leq 2R \int_0^{\pi/2} e^{-2aR\theta/\pi}\, d\theta = \frac{\pi}{a}\left(1 - e^{-aR}\right) \leq \frac{\pi}{a}. \tag{1.7}$$

Combining, we obtain the bound

$$\left| \int_{\mathcal{C}_R} f(z)\, dz \right| \le \frac{\pi}{a} M(R). \tag{1.8}$$

Taking $R \to \infty$: by assumption (1.1), $M(R) \to 0$, so the right-hand side of (1.8) vanishes by the squeeze theorem, and hence

$$\lim_{R\to\infty} \int_{\mathcal{C}_R} f(z)\, dz = 0. \tag{1.9}$$